and genotypic ratios are the same. There is no blending of genes. In the F_2 generation R^1 and R^2 genes segregate and recombine to produce red, pink and white in the ratio of 1 : 2 : 1. R^1 allele codes for an enzyme responsible for the formation of red pigment. R^2 allele codes for defective enzyme. R^1 and R^2 genotypes produce only enough red pigments to make the flower pink. Two R^1R^1 are needed for producing red flowers. Two R^2R^2 genes are needed for white flowers. If blending had taken place, the original pure traits would not have appeared and all F_2 plants would have pink flowers. It is very clear that Mendel's particulate inheritance takes place in this cross which is confirmed by the reappearance of original phenotype in F_2.

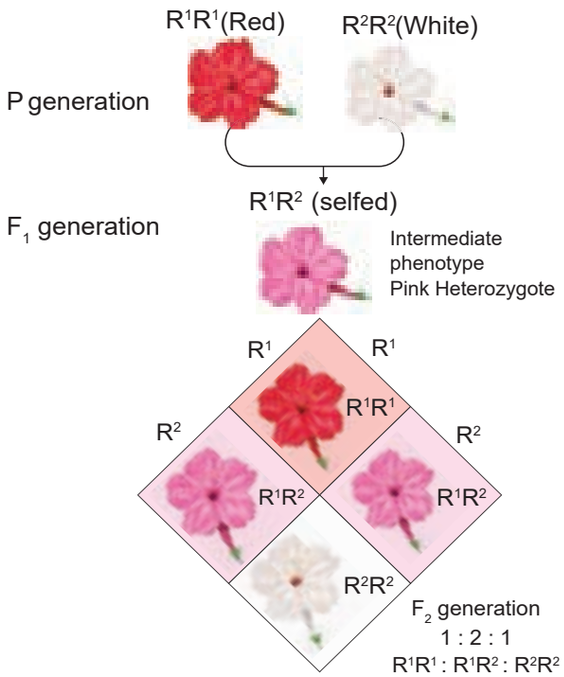


Figure 2.14: Incomplete dominance in 4 O' clock plant

2.4.2. Codominance (1 : 2 : 1)

This pattern occurs due to simultaneous (joint) expression of both alleles in the heterozygote - The phenomenon in which two alleles are both expressed in the heterozygous individual is known as codominance. Example: Red and white flowers of *Camellia*, inheritance of sickle cell haemoglobin, ABO blood group system in humanbeings. In humanbeings, I^A and I^B alleles of I gene are codominant which follows Mendels law of segregation. The codominance was demonstrated in plants with the help of electrophoresis or chromatography for protein or flavonoid substance. Example: *Gossypium hirsutum* and *Gossypium sturtianum*, their F_1 hybrid (amphiploid) was tested for seed proteins by electrophoresis. Both the parents have different banding patterns for their seed proteins. In hybrids, additive banding pattern was noticed. Their hybrid shows the presence of both the types of proteins similar to their parents.

The heterozygote genotype gives rise to a phenotype distinctly different from either of the homozygous genotypes. The F_1 heterozygotes produce a F_2 progeny in a phenotypic and genotypic ratios of 1 : 2 : 1.

How are we going to interpret the lack of dominance and give explanation to the intermediate heterozygote phenotype?

How will you explain incomplete dominance at the molecular level?

Gene expression is explained in a quantitative way. Wild-type allele which is a functional allele when present in two copies (R^1 R^1) produces an functional enzyme which synthesizes red pigments. The mutant allele which is a defective allele in two copies (R^2 R^2) produces an enzyme which cannot synthesize necessary red pigments. The white flower is due to the mutation causing complete loss of function. The F_1 intermediate phenotype heterozygote (R^1R^2) has one copy of the allele R^1. R^1 produces 50% of the functional protein resulting in half of the pigment of red flowered plant and so it is pink. The intermediate phenotype pink heterogyzote with 50% of functional protein is not enough to create the red phenotype homozygous, which makes 100% of the functional protein.

2.4.3. Lethal genes

An allele which has the potential to cause the death of an organism is called a "Lethal

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_033-052 CH 02.indd 41 3/14/2020 10:30:54 AM